Credit Risk Prediction Using Machine Learning

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Abstract

Credit risk prediction is one of the most important applications of machine learning in the financial industry. Banks and lending institutions need reliable methods to identify borrowers who are likely to default on loans. This project develops a machine learning framework for predicting loan default using borrower demographic, financial, and credit history information. Multiple machine learning algorithms were implemented and compared using the F1-score as the primary evaluation metric. The tuned XGBoost model achieved the best performance with an F1-score of 0.8122 and was selected as the final model.

Contents

1 Introduction

Credit risk refers to the possibility that a borrower may fail to repay a loan according to agreed conditions. Financial institutions must evaluate the risk associated with each loan application before approval.

Traditional methods often rely on manual assessment and predefined rules. However, machine learning techniques can automatically identify complex relationships in historical loan data and provide more accurate predictions.

The objective of this project is to build a machine learning model that predicts whether a borrower is likely to default on a loan.

2 Problem Statement

The aim of this project is to classify borrowers into two categories:

- Non-Default
- Default

using borrower demographic, financial, and credit history information.

3 Dataset Description

The dataset contains information about loan applicants and their financial history.

The features include:

- Age of borrower
- Annual income
- Home ownership status
- Employment length
- Loan purpose
- Loan grade
- Loan amount
- Interest rate
- Loan-to-income ratio
- Previous default history
- Credit history length

The target variable indicates whether a borrower defaulted on a loan.

4 Data Preprocessing

4.1 Handling Missing Values

Missing values were identified and treated appropriately.

- Missing employment length values were replaced using median imputation.
- Missing interest rate values were replaced using grade-wise median values.

4.2 Outlier Treatment

Outliers were carefully handled because extreme values can contain important financial information.

The following unrealistic observations were treated:

- Age greater than 80 years
- Employment length greater than 60 years

4.3 Log Transformation

Income distribution was positively skewed. A logarithmic transformation was applied to improve normality and reduce skewness.

4.4 Categorical Encoding

Categorical variables were transformed into numerical format using:

- Binary Encoding
- Ordinal Encoding
- One-Hot Encoding

4.5 Feature Scaling

Numerical variables were standardized before model training to ensure consistent scaling.

5 Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the characteristics of the dataset.

The following analyses were conducted:

- Missing value analysis
- Outlier analysis
- Distribution analysis
- Correlation analysis

- Loan default analysis
- Class distribution analysis

Several visualizations including histograms, boxplots, count plots, and heatmaps were generated to identify important patterns.

6 Feature Engineering

Feature engineering involved preparing the data for machine learning algorithms.

The following tasks were performed:

- Encoding categorical variables
- Removing multicollinearity
- Feature selection
- Standardization of numerical features

Variance Inflation Factor analysis was used to identify multicollinearity among predictors.

7 Machine Learning Models

The following machine learning algorithms were implemented and evaluated.

7.1 Decision Tree

Decision Trees learn hierarchical decision rules and can capture nonlinear relationships.

7.2 Support Vector Machine

Support Vector Machines construct optimal decision boundaries between classes.

7.3 Random Forest

Random Forest combines multiple decision trees and improves predictive stability.

7.4 XGBoost

XGBoost is a gradient boosting algorithm that iteratively improves prediction accuracy by minimizing classification errors.

8 Evaluation Metric

8.1 Why F1-Score Was Used

The dataset is imbalanced because non-default borrowers significantly outnumber default borrowers.

In such situations, accuracy can be misleading because a model may achieve high accuracy simply by predicting the majority class.

Therefore, the F1-score was selected as the primary evaluation metric.

The F1-score is defined as

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

The F1-score balances Precision and Recall and provides a more reliable measure of performance for imbalanced classification problems such as credit risk prediction.

A higher F1-score indicates better identification of risky borrowers while reducing misclassification errors.

9 Results

Table 1: Comparison of Models Using F1-Score

Model	F1 Score
Decision Tree	0.7245
Support Vector Machine	0.7288
Random Forest	0.6902
Tuned Random Forest	0.7892
XGBoost	0.7977
Tuned XGBoost	0.8122

10 Discussion

The Decision Tree model achieved an F1-score of 0.7245 and successfully captured nonlinear relationships in the data.

The Support Vector Machine achieved an F1-score of 0.7288 and slightly outperformed the Decision Tree model.

The baseline Random Forest model achieved an F1-score of 0.6902. After hyperparameter tuning, the performance improved substantially to an F1-score of 0.7892.

The baseline XGBoost model achieved an F1-score of 0.7977.

The tuned XGBoost model achieved the highest F1-score of 0.8122 among all evaluated models.

The superior performance of XGBoost can be attributed to:

- Gradient boosting framework
- Better handling of nonlinear relationships

- Strong regularization
- Effective learning of feature interactions

11 Final Model Selection

The tuned XGBoost model was selected as the final model because it achieved the highest F1-score.

Best hyperparameters:

- Number of estimators: 200
- Maximum depth: 7
- Learning rate: 0.2
- Subsample ratio: 1.0
- Feature sampling ratio: 0.8

Final F1-score:
[0.8122]

12 Conclusion

This project developed a machine learning framework for credit risk prediction using borrower information.

Multiple classification algorithms were evaluated and compared using F1-score as the primary evaluation metric.

Among all tested models, the tuned XGBoost model achieved the highest F1-score of 0.8122 and was selected as the final model.

The results demonstrate that advanced boosting algorithms can effectively identify high-risk borrowers and support better lending decisions in financial institutions.

13 Future Work

Future improvements may include:

- Handling class imbalance using SMOTE
- Explainable AI using SHAP values
- Deployment through Streamlit
- Comparison with LightGBM and CatBoost
- Real-time credit risk prediction systems

14 References

1. Scikit-Learn Documentation.
2. XGBoost Documentation.
3. Christopher Bishop, Pattern Recognition and Machine Learning.
4. Credit Risk Modeling Literature.